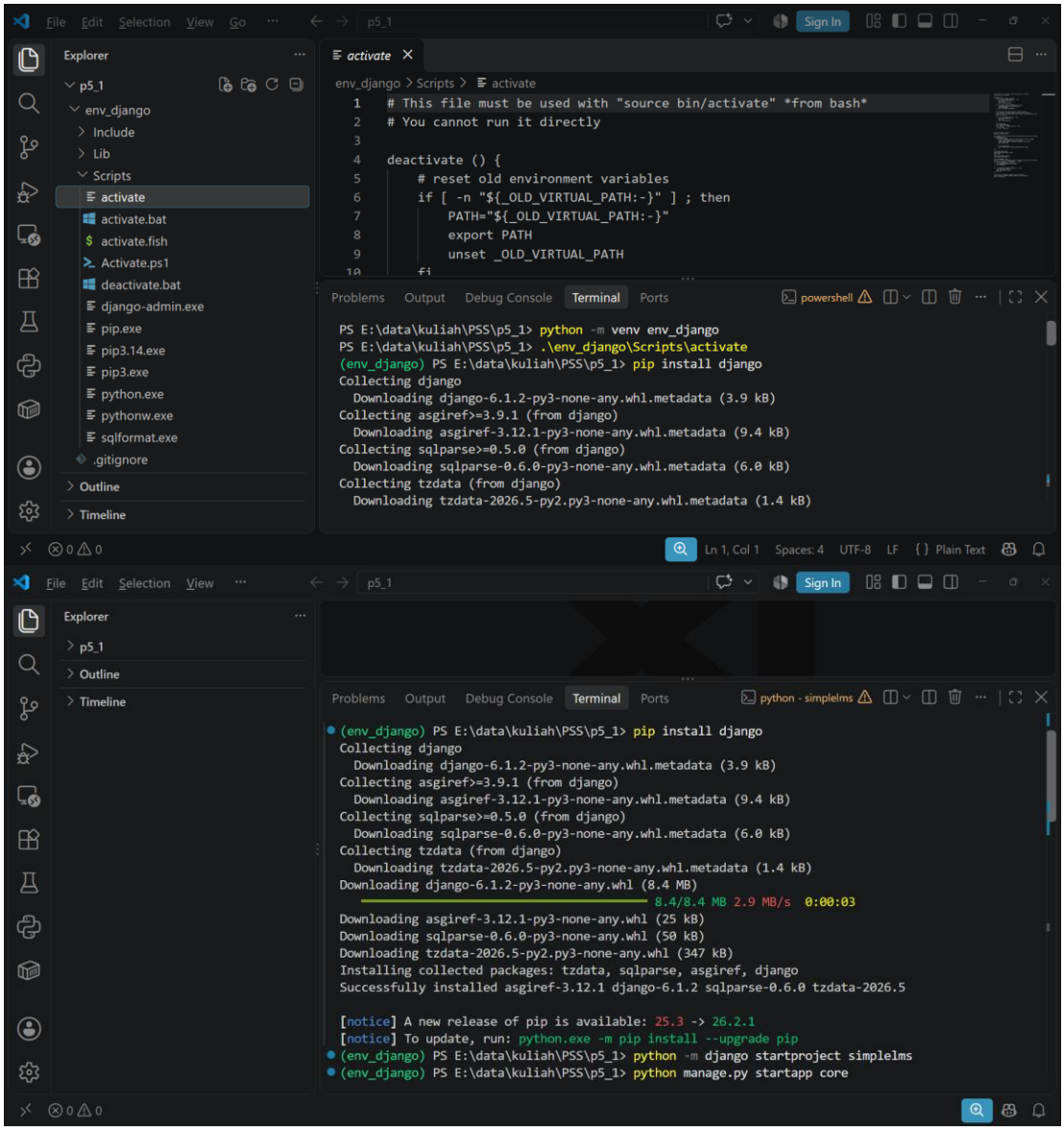
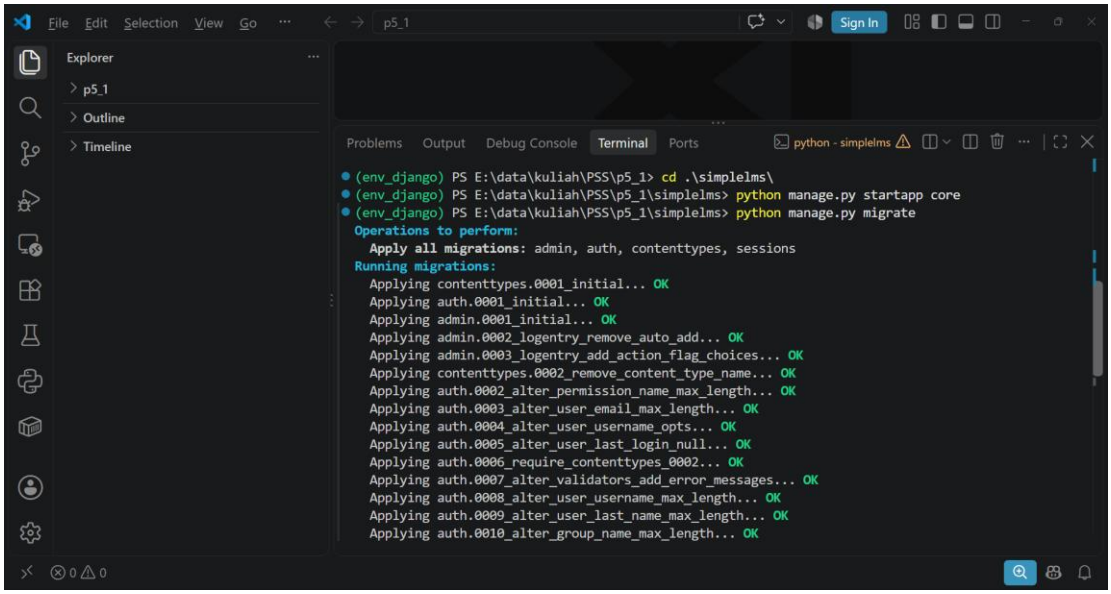
Pertemuan 5 : Aplikasi Django Menggunakan Docker

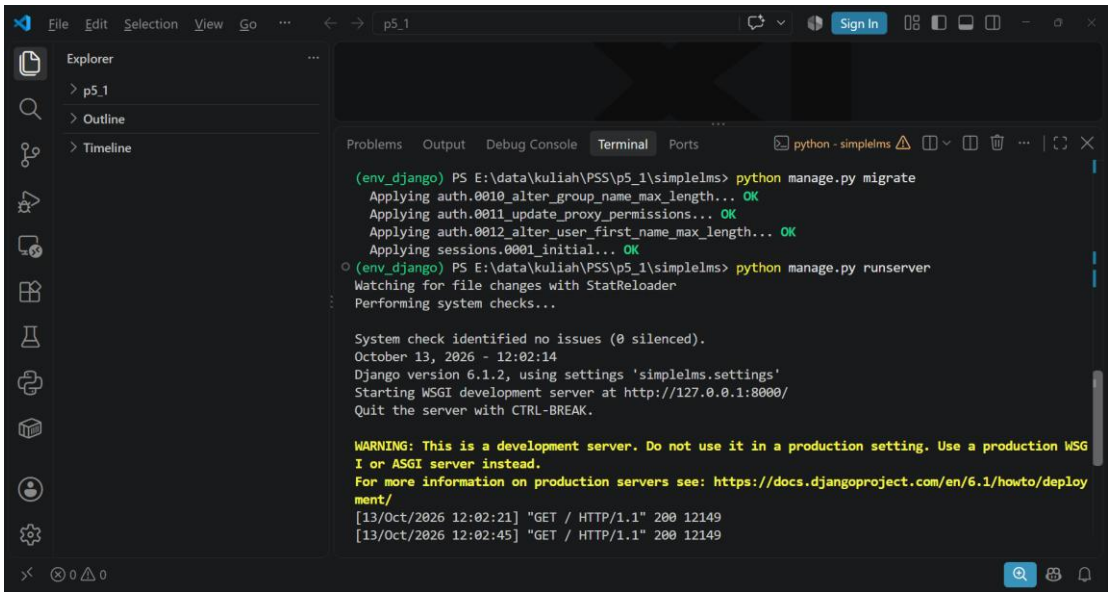
1. Deploy Aplikasi Django Menggunakan Docker
   Studi kasus : Simple_LMS contoh di materi 5.

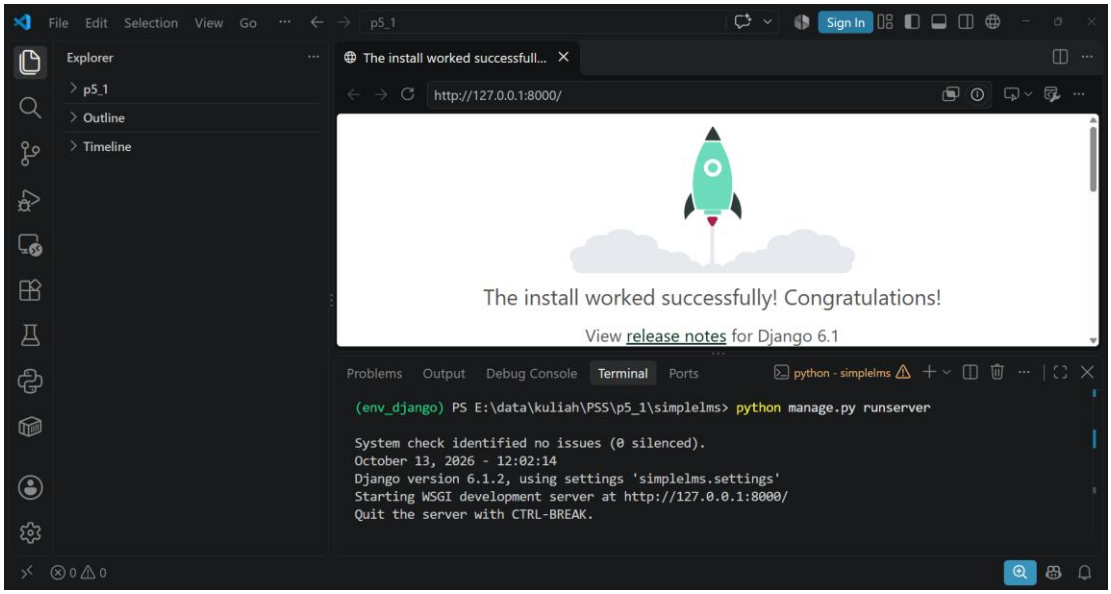
2. Deploy Aplikasi Django Menggunakan Docker To-Do-List dengan database PostgreSQL.

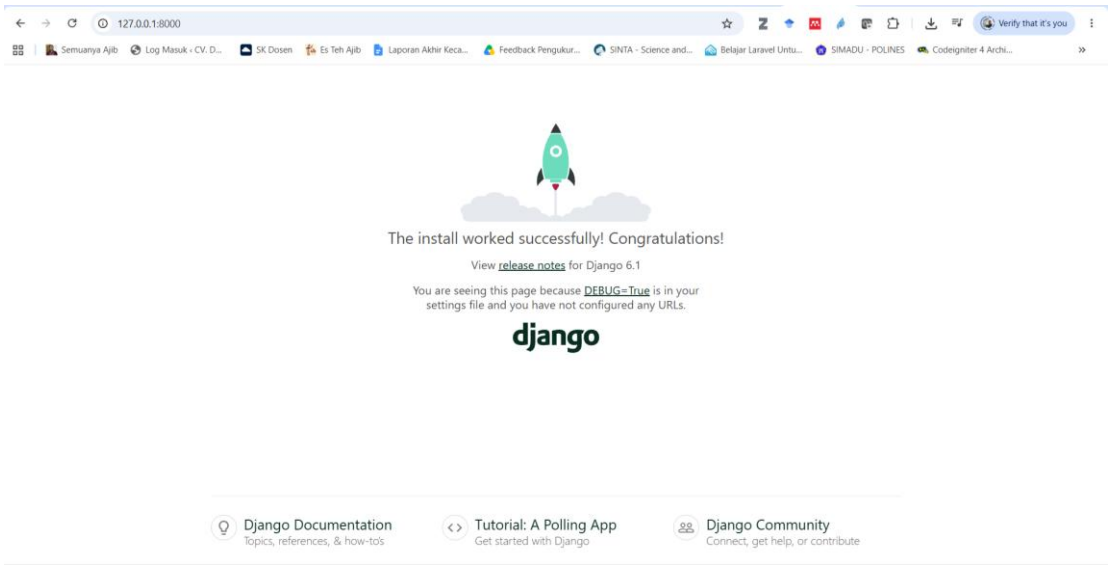
Django dengan Python



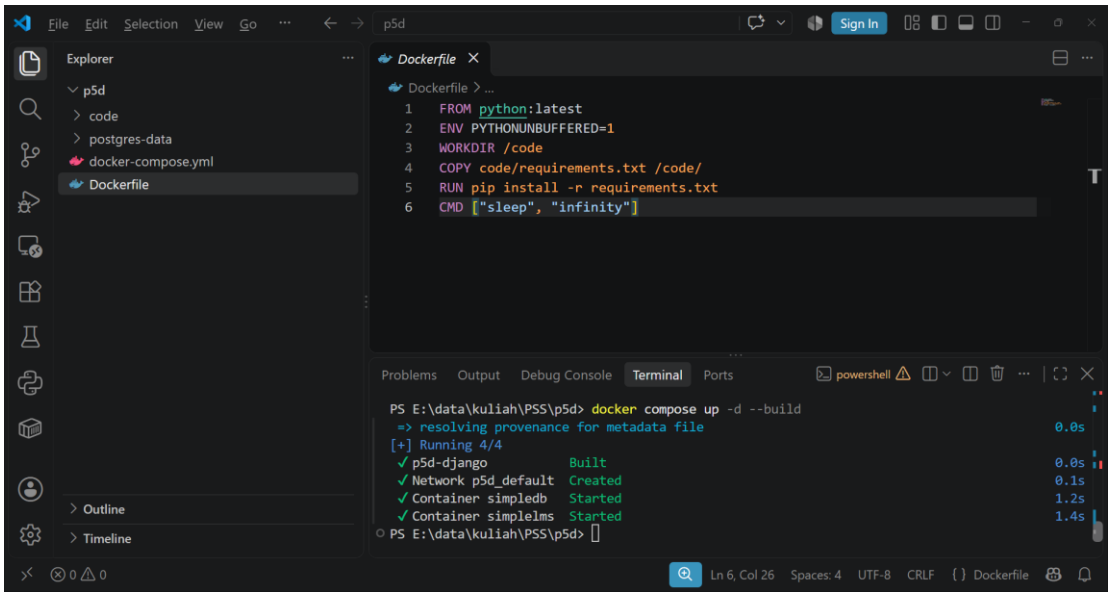

File Edit Selection View Go
p5_1
Sign In
Explorer
p5_1
Outline
Timeline
Problems Output Debug Console Terminal Ports
python - simplelms
(env_django) PS E:\data\kuliah\PSS\p5_1> cd .\simplelms\
(env_django) PS E:\data\kuliah\PSS\p5_1\simplelms> python manage.py startapp core
(env_django) PS E:\data\kuliah\PSS\p5_1\simplelms> python manage.py migrate
Operations to perform:
Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
Applying contenttypes.0001_initial... OK
Applying auth.0001_initial... OK
Applying admin.0001_initial... OK
Applying admin.0002_logentry_remove_auto_add... OK
Applying admin.0003_logentry_add_action_flag_choices... OK
Applying contenttypes.0002_remove_content_type_name... OK
Applying auth.0002_alter_permission_name_max_length... OK
Applying auth.0003_alter_user_email_max_length... OK
Applying auth.0004_alter_user_username_opts... OK
Applying auth.0005_alter_user_last_login_null... OK
Applying auth.0006_require_contenttypes_0002... OK
Applying auth.0007_alter_validators_add_error_messages... OK
Applying auth.0008_alter_user_username_max_length... OK
Applying auth.0009_alter_user_last_name_max_length... OK
Applying auth.0010_alter_group_name_max_length... OK


File Edit Selection View Go
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Sign In
Explorer
p5_1
Outline
Timeline
Problems Output Debug Console Terminal Ports
python - simplelms
(env_django) PS E:\data\kuliah\PSS\p5_1\simplelms> python manage.py migrate
Applying auth.0010_alter_group_name_max_length... OK
Applying auth.0011_update_proxy_permissions... OK
Applying auth.0012_alter_user_first_name_max_length... OK
Applying sessions.0001_initial... OK
(env_django) PS E:\data\kuliah\PSS\p5_1\simplelms> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...
System check identified no issues (0 silenced).
October 13, 2026 - 12:02:14
Django version 6.1.2, using settings 'simplelms.settings'
Starting WSGI development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/6.1/howto/deployment/
[13/Oct/2026 12:02:21] "GET / HTTP/1.1" 200 12149
[13/Oct/2026 12:02:45] "GET / HTTP/1.1" 200 12149


File Edit Selection View Go
p5_1
Sign In
Explorer
p5_1
Outline
Timeline
The install worked successfull...
http://127.0.0.1:8000/
The install worked successfully! Congratulations!
View release notes for Django 6.1
Problems Output Debug Console Terminal Ports
python - simplelms
(env_django) PS E:\data\kuliah\PSS\p5_1\simplelms> python manage.py runserver
System check identified no issues (0 silenced).
October 13, 2026 - 12:02:14
Django version 6.1.2, using settings 'simplelms.settings'
Starting WSGI development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.


127.0.0.1:8000
The install worked successfully! Congratulations!
View release notes for Django 6.1
You are seeing this page because DEBUG=True is in your settings file and you have not configured any URLs.
django
Django Documentation
Topics, references, & how-to's
Tutorial: A Polling App
Get started with Django
Django Community
Connect, get help, or contribute

Docker Django

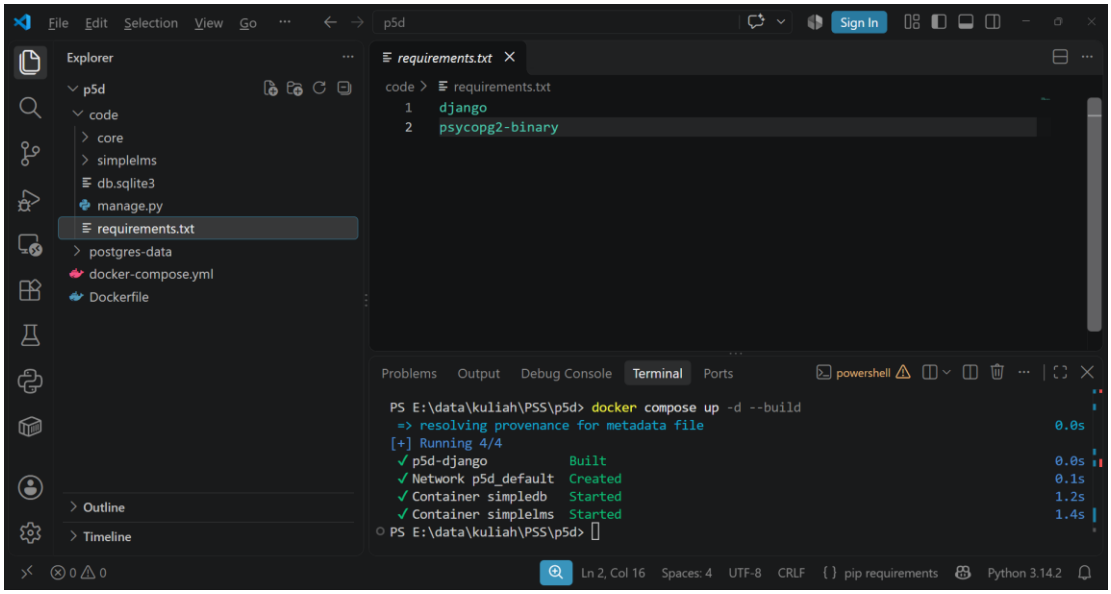


Dockerfile :

```
FROM python:latest
ENV PYTHONUNBUFFERED=1
WORKDIR /code
COPY code/requirements.txt /code/
RUN pip install -r requirements.txt
CMD ["sleep", "infinity"]
```

code/requirements.txt :

```
django
psycopg2-binary
```



docker-compose.yml

```
services:
  django:
    container_name: simplelms
    build: .
    #command: sleep infinity
    command: python manage.py runserver 0.0.0.0:8000
    volumes:
      - ./code:/code
    ports:
      - "8007:8000"
    depends_on:
    - postgres
  postgres:
    container_name: simpledb
    image: postgres:17
    environment:
      POSTGRES_DB: simple_lms
      POSTGRES_USER: simple_user
      POSTGRES_PASSWORD: simple_password
      PGDATA: /var/lib/postgresql/data/pgdata
    ports:
      - "5532:5432"
    volumes:
      - ./postgres-data:/var/lib/postgresql/data/pgdata
```

Jalankan :

```
docker compose up -d --build
```

Membuat project simplelms :

```
docker exec simplelms django-admin startproject simplelms .
docker exec simplelms python manage.py startapp core
```

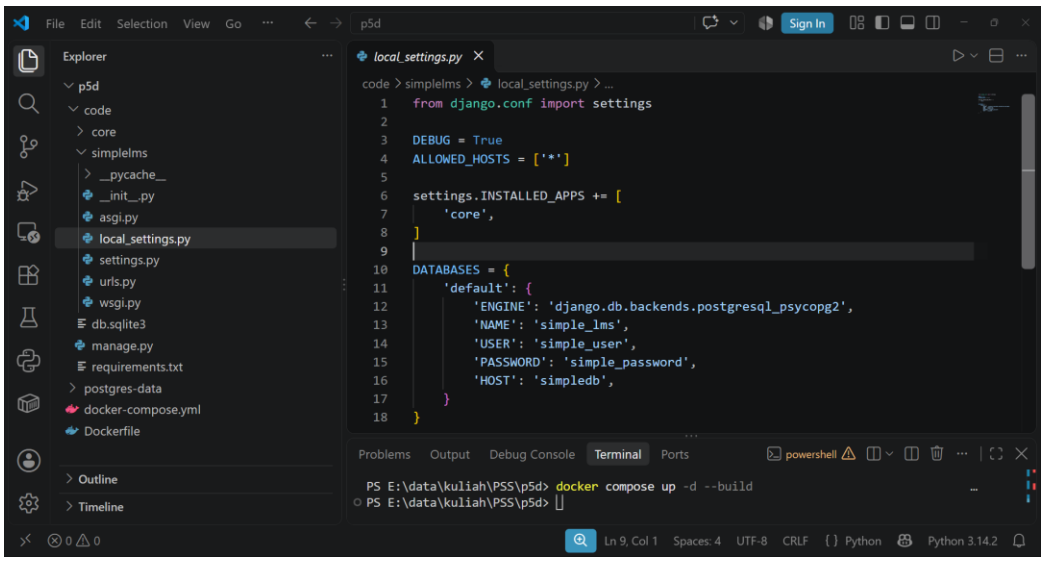
Buat simplelms/local_settings.py :

```
from django.conf import settings

DEBUG = True
ALLOWED_HOSTS = ['*']

settings.INSTALLED_APPS += [
    'core',
]

DATABASES = {
    'default': {
       'ENGINE': 'django.db.backends.postgresql_psycopg2',
        'NAME': 'simple_lms',
        'USER': 'simple_user',
        'PASSWORD': 'simple_password',
        'HOST': 'simpledb',
    }
}
```

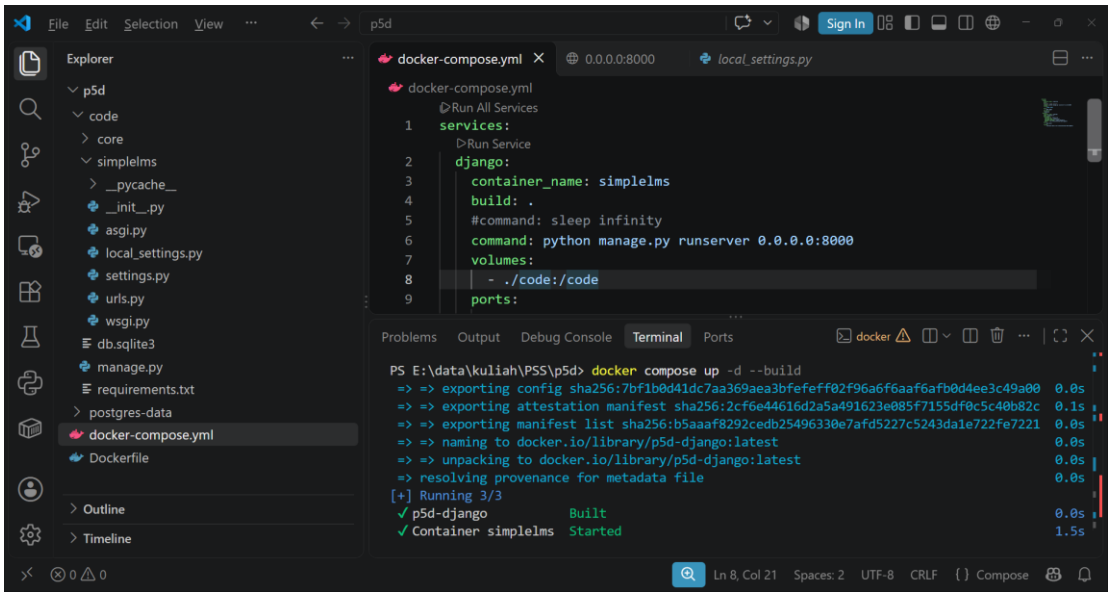


Migrasi Basis Data :

```
docker exec simplelms python manage.py makemigrations
docker exec simplelms python manage.py migrate
```

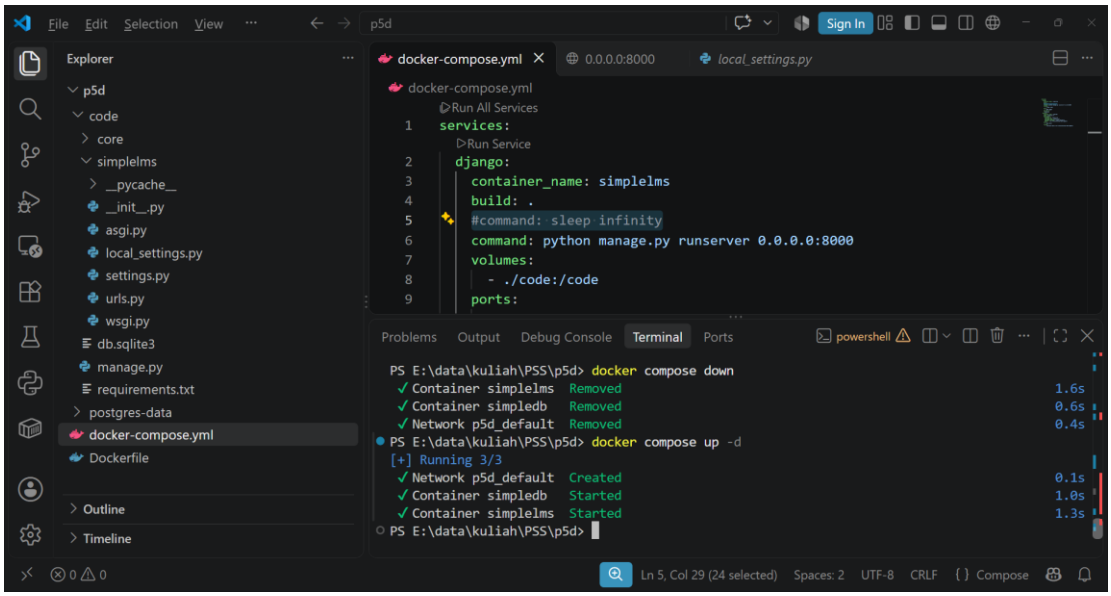
Ubah di docker-compose.yml :

```
#command: sleep infinity
command: python manage.py runserver 0.0.0.0:8000
```

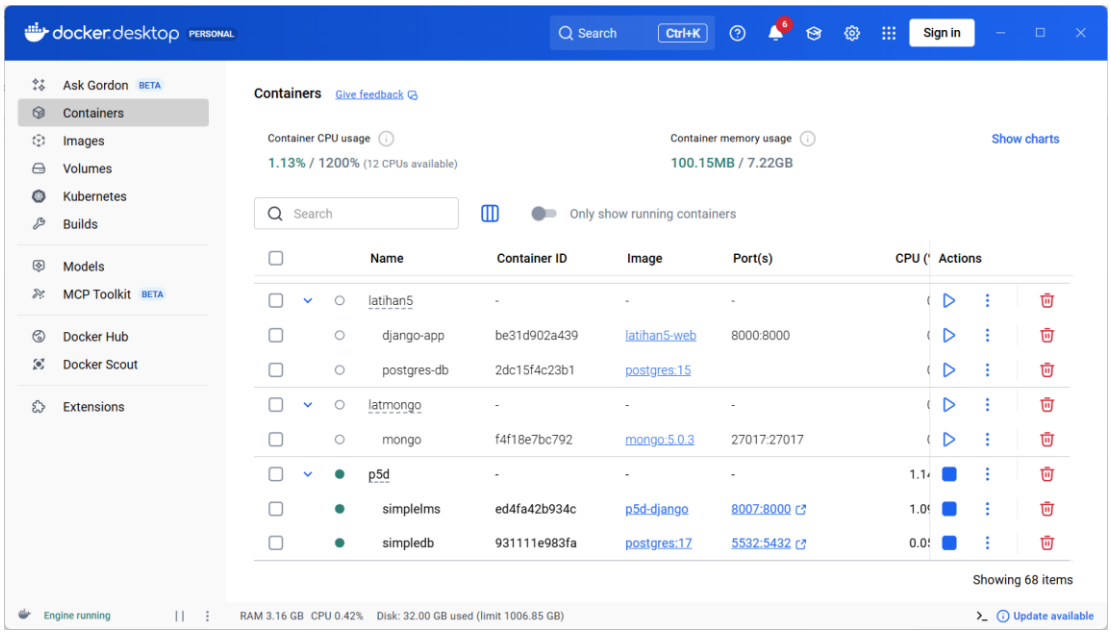


Down Up Container :

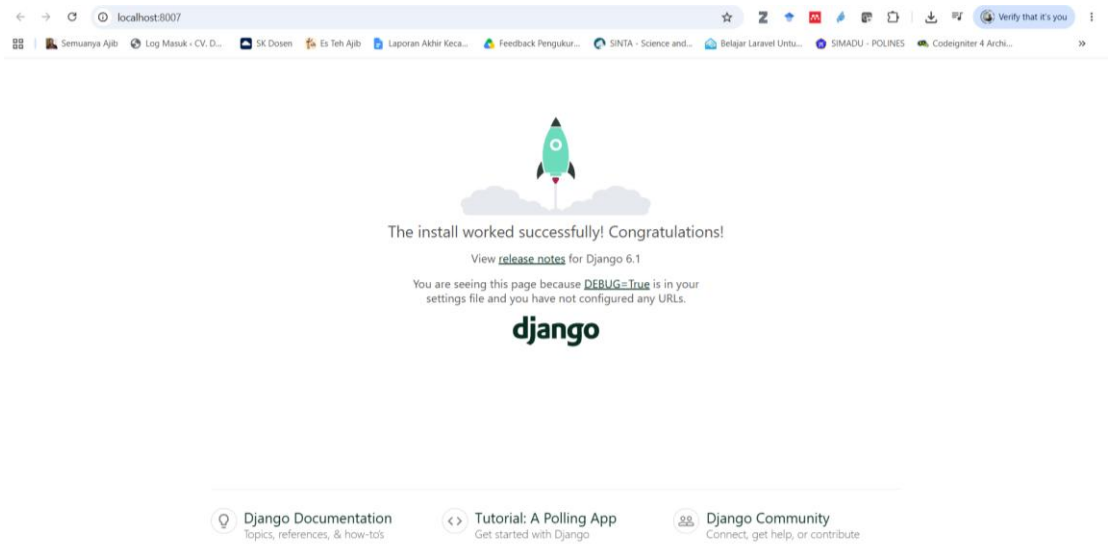
```
docker compose down
docker compose up -d
```



Buka di Docker :



Test simplelms :



Membuat superuser :

```
docker exec -it simplelms python manage.py createsuperuser
```

```
PS E:\data\kuliah\PSS\p5d> docker exec -it simplelms python manage.py createsuperuser
Username (leave blank to use 'root'): ajib
Email address: ajibsusanto@gmail.com
Password:
Password (again):
```

Jalankan halaman admin di browser :

`localhost:8007/admin`

